THE EVOLUTION OF 3-POINT SHOOTING IN THE NBA

Jose Luis Ganfornina Mata

Introduced as an experimental rule in the 1979-80 NBA season, the three-point shot has evolved into the defining offensive strategy of modern NBA basketball.

Full Project

SUMMARY

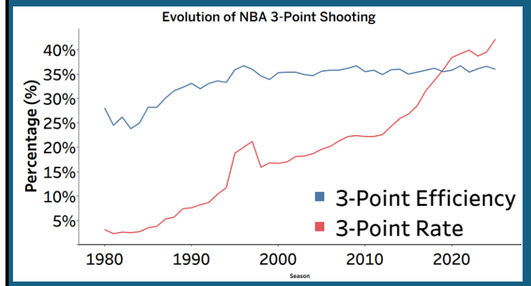
This project analyzes the evolution of three-point shooting in the NBA since the 1979–80 season, concentrating on trends in usage, efficiency, and player roles, with particular attention to how Centers have adjusted.

DATA & METHODOLOGY

Datasets from **Kaggle's** "NBA Stats (1947-present)" were used, focusing on **team** and **player** season totals from **1980 onwards**. Duplicate player records were combined, and null values set to zero. Data was aggregated by team, player, and season, analysing games played, shot attempts, and player position.

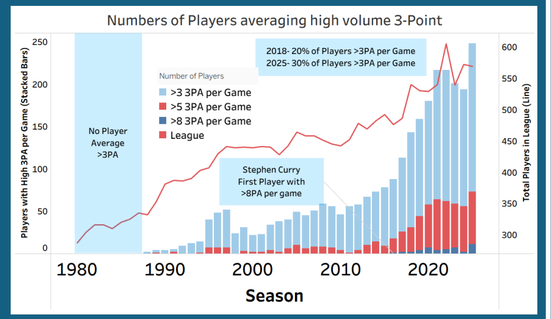
SQL was used to compute three-point metrics and trends, while **Tableau** was used for visualization.

LEAGUE EVOLUTION

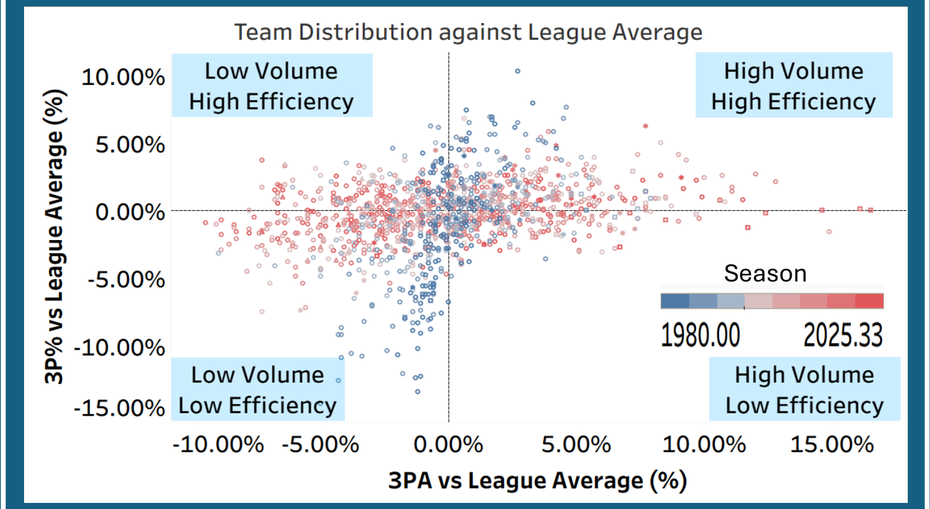


Three-point attempts increased dramatically since the 1980's, while efficiency stabilized around ~35%, suggesting teams gained offensive value primarily through increased shot volume rather than major improvements in efficiency.

The number of players attempting large volumes of three-pointers per game has grown significantly since the mid-2010s.



TEAM SHOOTING STRATEGIES



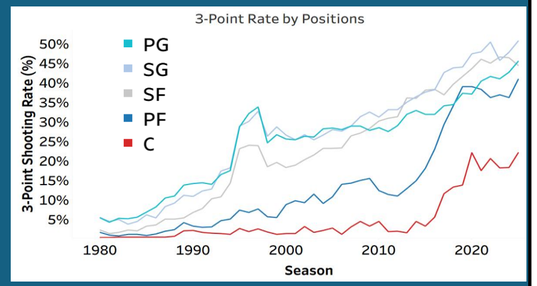
Over time, modern teams increasingly shift toward **high-volume three-point strategies**, reflecting the growing importance of the three-point shot.

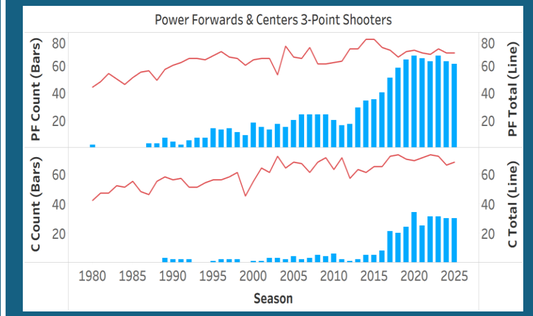
For a more detailed exploration, please visit the interactive **DASHBOARD**

Click or Scan

PLAYER ROLES

Three-point shooting has expanded across all positions, with Power Forwards and Centers showing the fastest growth since the mid-2010s.





The number of Power Forwards and Centers attempting at least one three-point per game has increased considerably in the last decade.

CONCLUSIONS

- Three-point shooting has evolved from a specialized tactic into a fundamental component of offensive strategy.
- Teams now place greater emphasis on **increasing shot volume rather than focusing solely on marginal efficiency improvements**.
- The **mid-2010s witnessed a significant strategic shift** towards prioritising perimeter shooting.
- The **Golden State Warriors**, led by **Stephen Curry**, were among the pioneers in combining high-volume three-point attempts with exceptional efficiency.
- Three-point offensive strategies have reshaped modern basketball, requiring players in every position on the court to adjust accordingly.

LIMITATIONS & FUTURE WORK

- **Team-level analysis may mask player-level variation**, as different players within the same team may have distinct shooting roles and responsibilities.
- **Position classifications are simplified;** modern basketball increasingly features hybrid and positionless roles.
- The analysis focuses primarily on **three-point volume and efficiency**, without other contextual factors such as pace of play, defensive schemes, or rule changes.
- **Future work could incorporate shot location and tracking data** to better analyze spacing, shot selection, and offensive strategies.
- Additional analysis could explore **player-level trends and career development** in three-point shooting.